The University of Melbourne
Semester 2 Assessment 2015

School of Physics

PHYC10004
PHYSICAL SCIENCE AND TECHNOLOGY

Reading time: **15 minutes**
Exam Duration: **3 hours**
This paper has: **8 pages + 2 pages of formula**

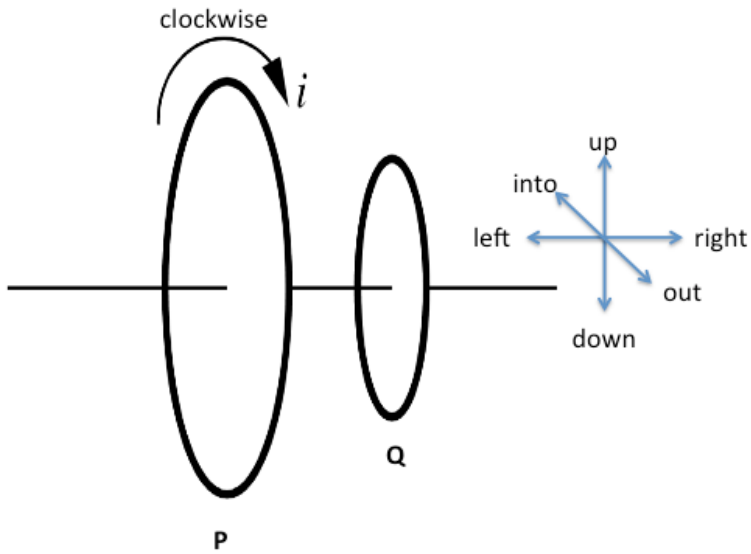
Authorised Materials:	All calculators are permitted, but should not have been programmed nor should they store additional information. Dictionaries, including language dictionaries, are prohibited.

Instructions to Invigilators:	The final two pages of this exam paper contain formulae and data for the use of candidates. These pages may be detached. **This examination paper is to remain in the examination room. Students may not retain any part of this exam.**

Instructions to students:	Attempt ALL questions. The total number of marks is 150. Two pages containing formulae and data are included at the end of the exam paper. These pages may be detached, but may not be removed from the examination room.

Paper to be held by the Baillieu Library.

Question 1

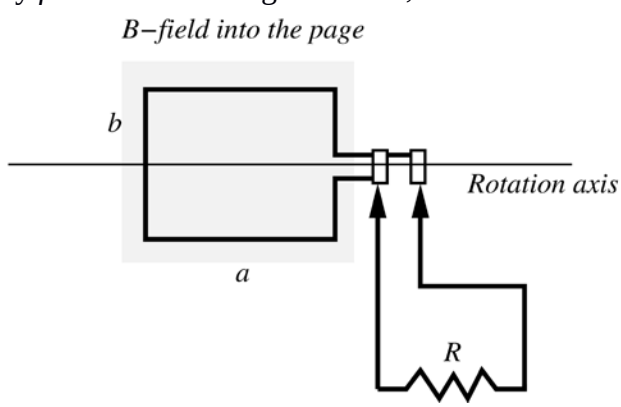


(a) Two conducting loops labelled P and Q are arranged as shown above. A clockwise current, viewed from the left, *i*, is suddenly established in the left-hand loop P, as indicated in the diagram. Use the key at the right of the diagram to describe directions.

i. What is the direction of the current induced in Q (the right-hand loop)?

ii. At the instant the current is established in P, does the right-hand loop experience a force and, if so, in what direction is that force? Explain your reasoning.

(b) A wire is formed into a rectangular coil consisting of *N* turns, with each turn being of length *a* and width *b*. As shown in the figure above, this coil is connected to a resistor, *R*, in such a way that the coil can be rotated at frequency *f* in a uniform magnetic field, **B**.



i. Find an expression for the flux through one turn of the loop as a function of time.

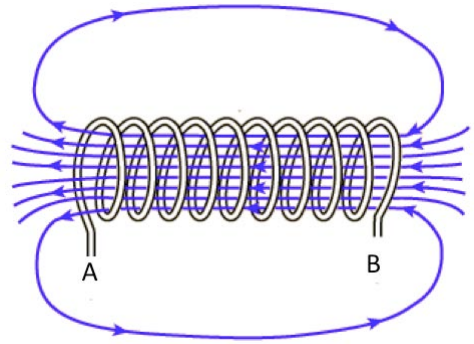
ii. Show that the induced EMF, $E(t)$, can be given by $(t) = 2\pi fN(ab)B\sin(2\pi ft)$ where $B = |\mathbf{B}|$.

iii. Find an expression for the power dissipated in the resistor.

(2 + 3 + 3 + 4 + 3 = 15 marks)

Question 2

A 20 m length of copper wire has been wound into a solenoid of 1000 turns, as shown in the diagram. (The diagram is not to scale.) A current of 2 A flows through this wire, producing a magnetic field as shown.



(a) Consider a small section inside the solenoid.

i. Use Ampère's Law to show that the magnitude of the magnetic field in the interior of this long solenoid is given by $B = \mu_o n i$, where μ_o is the magnetic permeability, i is the current in the wire coil that forms the solenoid and n is the number of turns of wire per metre. Clearly explain in your answer any assumptions that you make.

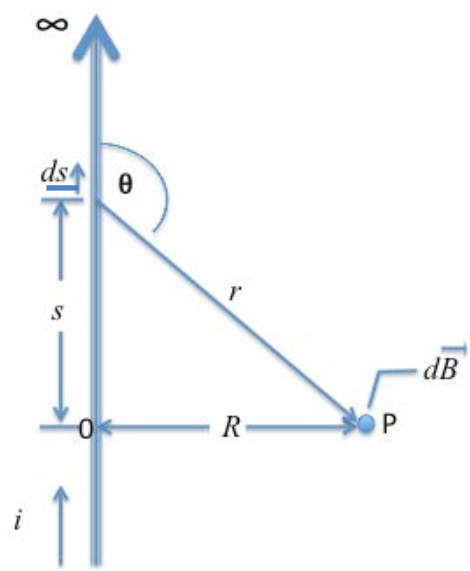
ii. In which direction is the current flowing : A→ B or B→ A ?

iii. Calculate the strength of the magnetic field produced inside this solenoid.

(b) Consider an infinite straight wire, directed along the z-axis carrying a current i, as shown in the diagram. The magnitude of the differential magnetic field produced at P by the current-length element $i\,d\vec{s}$ located a distance **r** from P is given by the Biot-Savart Law:

$$dB = \frac{\mu_o}{4\pi}\frac{i\,ds\sin(\theta)}{r^2}$$

Apply the above to show that the magnitude of the magnetic field B, generated by the current i is given by $B = \frac{\mu_o i}{2\pi R}$, where $\boldsymbol{R}$ is the perpendicular distance from the wire to the point at which $\boldsymbol{B}$ is measured.

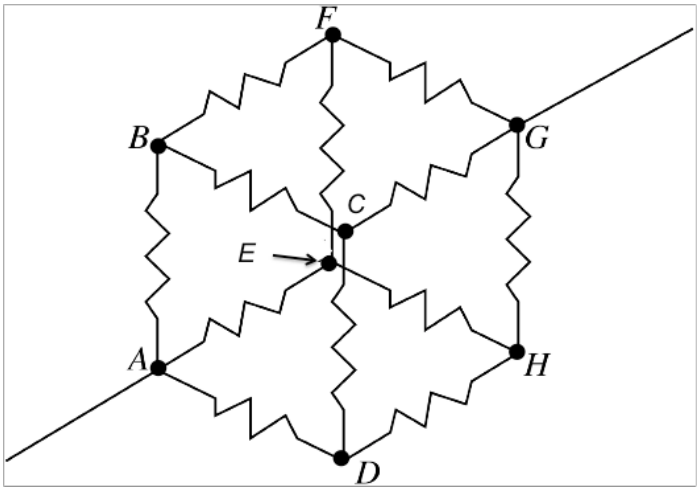


You may find the following result useful:

$$\int \frac{ds}{(s^2+R^2)^{\frac{3}{2}}} = \frac{s}{R^2(s^2+R^2)^{\frac{1}{2}}} + C$$

(4 + 1 + 2 + 8 = 15 marks)

Question 3



Twelve resistors, each of resistance R, are connected to form the edges of a cube, as shown above.

(a) An EMF is connected to this cube at points A and G. Identify which vertices (labelled A-H in the diagram) are at the same potential.

(b) By connecting vertices that share the same potential show that the network can be replaced by three resistors connected in series, where each of these resistors is constructed from a number of the original resistors, R, connected in parallel.

(c) Hence or otherwise how that the resistor cube can be replaced by a single equivalent resistor whose resistance is $5R/6$.

(3 + 4 + 5 = 12 marks)

Question 4

(a) A point positive charge, $+Q$, is placed at the centre of a spherical cavity of radius R formed in a large block of conducting material. Use Gauss' Law to find the magnitude and direction of the electric field at a distance

i $r = R/2$

ii $r = R$

iii $r >$ R from the centre of the cavity.

(b) Two long concentric conducting cylinders of radius a and b, respectively, each carry equal and opposite charges, with a linear charge density λ C m^{-1}. The cylinders are of negligible thickness.

Assuming that $b > a$, use Gauss's law

i. to show that $\boldsymbol{E}$ vanishes for $r < a$ and $r > b$

ii. to show that for $a < r < b$, $|E| = \dfrac{\lambda}{2\pi\varepsilon_0 r}$

iii. to describe the direction of the electric field in the space between the cylinders

(9 + 3 + 4 + 2 = 18 marks)

Question 5

(a)

i Identify the 'displacement current' in Maxwell-Ampere's law.

ii In what situations would you expect to find a non-zero displacement current?

iii Do displacement currents involve the flow of charge?

(b) A displacement current, i_d, exists in a circular region of radius R; the direction of i_d is out of the page. The magnitude of the displacement current is given by $i_d = i_0 (r/R)$, where i_0 is a real positive quantity and r is the radial distance from the centre of the region, so that $r \leq R$.

i What is the magnitude of the magnetic field due to the displacement current at $r = R/2$?

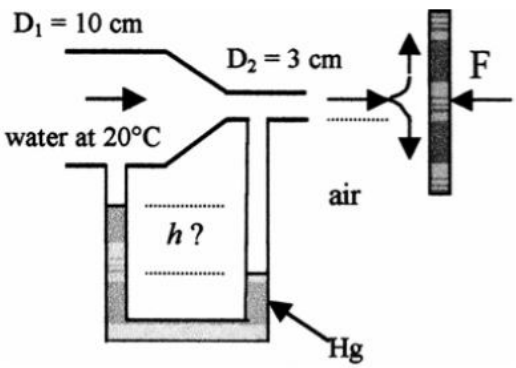
ii What is the magnitude of the magnetic field due to the displacement current at $r = 2R$?

iii Describe the shape and direction of the magnetic field due to this current.

(2 + 2 + 2 + 3 + 3+ 3 = 15 marks)

Question 6

Water flows through a circular nozzle, exits into the air as a jet, and strikes the plate F, as shown in the diagram. The two section of the nozzle have diameter D_1 = 10 cm and D_2 = 3 cm respectively.



The force required to hold the plate in place is 70 N. Assume frictionless one-dimensional flow and water density as 1000 kg m^{-3}.

i) If the exit speed of water from the tube of diameter 3 cm is v_2, show that the mass of water leaving the tube each second is given by $\rho A_2 v_2$, where $A_2 = (\pi/4)D_2^2$.

ii) Hence show that v_2 = 9.95 m/s. Show your working clearly.

iii) Calculate the velocity of the water at the entrance to the circular nozzle.

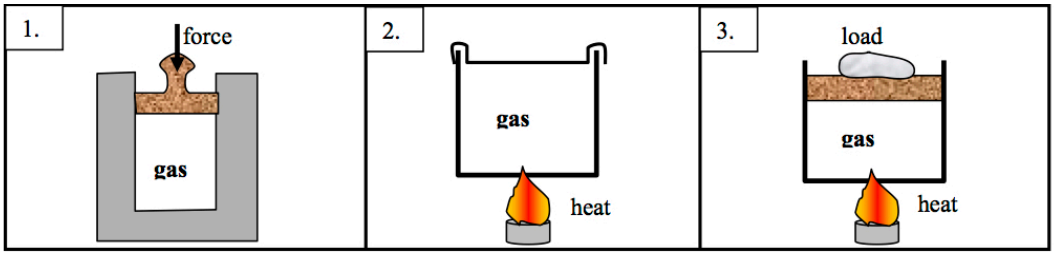
iv) Calculate the pressure difference measured by the manometer.

v) Calculate the mercury manometer reading h. Take the density of Hg to be 13 600 kg m^{-3}.

(3 + 2 + 2 + 2 + 1 = 10 marks)

Question 7

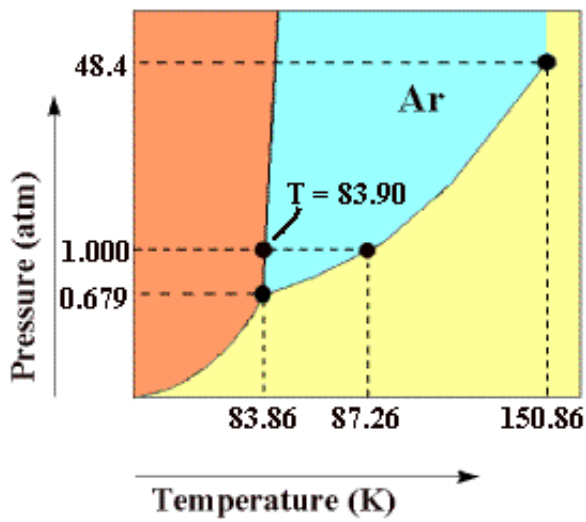
There are three systems depicted below. In the first system, the piston is pushed down into an insulated cylinder. In the second system, the gas inside the tin is heated with the lid firmly in place. In the third system the gas is heated in a container with a freely sliding lid on with a load on top.



(a)

i. Are any of these processes adiabatic? If so, identify which one(s).
ii. Are any of these processes isochoric? If so, identify which one(s).
iii. Are any of these processes isobaric? If so, identify which one(s).

Argon has normal melting and boiling points of 83.90 and 87.26 K; its triple point is at 83.86 K and 0.679 atm, and its critical temperature and pressure are 150.86 K and 48.4 atm, as shown.



(from http://www.webassign.net/; not to scale)

b) State whether Argon is a solid, liquid, or gas under each of the following conditions:

i 1.000 atm and 90 K
ii 0.685 atm and 80 K

c) 30 g of ideal gas are contained in a cylinder fitted with a freely sliding piston.
Its temperature is 15°C. The piston is moved with an average force of 200 N so that that it moves 60 mm and compresses the gas.
The temperature rises to 21°C as a result.
Calculate the magnitude of the heat transfer involved, given c_v= 718 J kg^{-1} K^{-1}.

d) In a constant volume process 209 J of energy are transferred by heat to 1 mol of Helium gas (a monoatomic gas) that is initially at 300 K. Find:

i) the increase in the He internal energy
ii) the work done on the gas

(1 + 1 + 1 + 1 + 1 + 5 + 1 + 1 = 12 marks)

Question 8

a) An isolated copper sphere of radius 5 cm, initially uncharged, is illuminated with by ultraviolet light of wavelength 200 nm. Knowing that the work function for copper is 4.7 eV, calculate the maximum kinetic energy of the ejected electrons.

b) X-rays with an energy of 300 keV undergo Compton scattering from a target. The scattered rays are detected at 37° relative to the incident rays. Find:
   i. The Compton shift at this angle.
   ii. The energy of the scattered x-ray.
   iii. The energy of the recoiling electron.
   iv. The ratio of the photon wavelength to the electron wavelength.

c) After learning about matter waves, an 80 kg student has concerns about being diffracted when passing through a doorway of width 75 cm. Assume that significant diffraction occurs when the width of the diffraction aperture is less than 10.0 times the wavelength of the wave being diffracted.
   i. Determine the maximum speed at which the student can pass through the doorway so as to be significantly diffracted.
   ii. With that speed, how long will it take the student to pass through the doorway if it is in a wall 15.0 cm thick?
   iii. Compare your result to the currently accepted age of the Universe (4×10^{17}s). Should this student worry about being diffracted?

(2 + 2 + 2 + 2 + 2 + 3 + 3 + 1 = 17 marks)

Question 9

A solid iron sphere has a 2.00 cm radius. Its temperature is uniform throughout its volume. It acts as an ideal blackbody. The density of iron is 7874 kg/m^3.

(a)
   i. Find the mass of the sphere.
   ii. Assume that it has a temperature of 20.0° C, find the power with which it is radiating electromagnetic waves.
   iii. Find the wavelength λ_{max}.

(b) Although the sphere emits a spectrum of waves all having different wavelengths, model its whole power output as carried by photons of wavelength λ_{max}:
   i. Find the energy of one photon.
   ii. Find the number of photons it emits each second.

(2 + 2 + 2 + 2 + 2 = 10 marks)

Question 10

a) An electron is confined between two impenetrable walls 0.2 nm apart. Determine the energy levels for the states n =1, 2 and 3.

b) A hydrogen atom makes a transition from the $n = 2$ to the $n = 1$ energy level. Find the wavelength and frequency of the emitted photon.

c) In interstellar space, highly excited hydrogen atoms, called Rydberg atoms, have been observed. These can be described by standard Bohr atom theory.
   i) Find the wavelength to which radio astronomers must tune to detect signals from electrons dropping from the $n = 273$ level to the $n = 270$ level.
   ii) What is the radius of the Rydberg atom for which $n = 273$?
   iii) What is the speed of an electron in this Rydberg atom?

d) When an atom makes a transition between states, energy is emitted in the form of a photon. Although an excited atom can radiate at any time from $t = 0$ to $t = \infty$, the average time interval after excitation during which an atom radiates is called the lifetime. If the lifetime is 1.0×10^{-8} s, use the uncertainty principle to compute the line width Δf produced by this finite lifetime.

(3 + 3 + 3 + 2 + 2 + 3 = 16 marks)

Question 11

In 1991 a German tourist discovered the well-preserved remains of a human, later dubbed "Ice Man," trapped in a glacier in the Italian Alps. Radioactive dating of a sample of Ice Man established an age of 5300 yr.

a) Why did scientists date the sample with the isotope C^{14}?

b) Find the radius of a ${}^{14}_{6}C$ nucleus.

c) Calculate the decay constant for ${}^{14}C$, which has a half-life of 5730 years.

d) Scientists used a 25.0 g piece of tissue of the Ice Man. The sample shows a ${}^{14}C$ activity of 250 decays/min. How many atoms of ${}^{14}C$ are in the sample?

e) Calculate how many ${}^{14}C$ nuclei were in the sample before decay.

(2 + 2 + 2 + 2 + 2 = 10 marks)

END OF EXAMINATION QUESTIONS